

Experiment: 9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim

To analyze the performance of a Massive MIMO system by studying channel capacity,

signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objectives

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR)

values using MATLAB. Observe how channel capacity increases with improving signal quality

based on the Shannon capacity principle.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB.

Analyze the effect of SNR on the quality and reliability of wireless communication in Massive

MIMO systems.

Objective – 3: Analyze Antenna Gain (dB)

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive

MIMO system using MATLAB. Observe how increasing the antenna array size improves

directional gain and enhances communication performance.

Objective1:

Program:

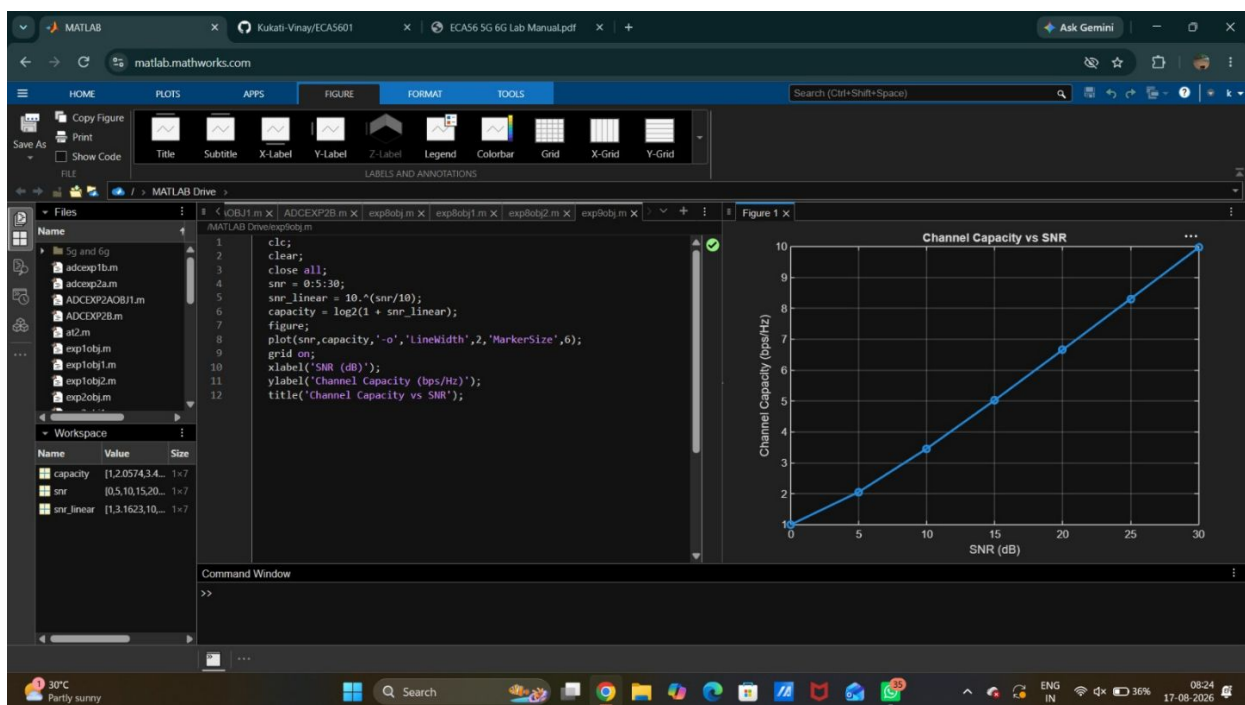
```
clc;
```

```
clear;
```

```

close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
output:

```



Objective2:

Program:

```

clc;
clear;

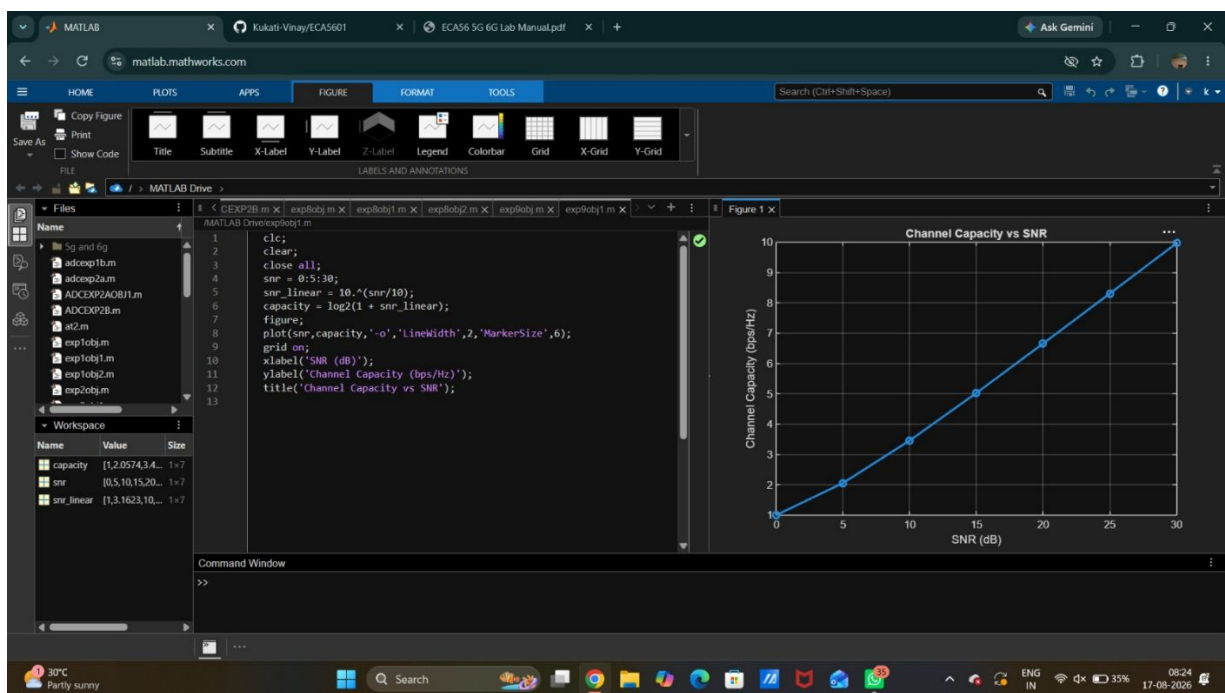
```

```

close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');

```

output:



Objective3:

Program:

```

clc;
clear;
close all;

```

```
antennas = [4 8 16 32 64]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Antenna Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

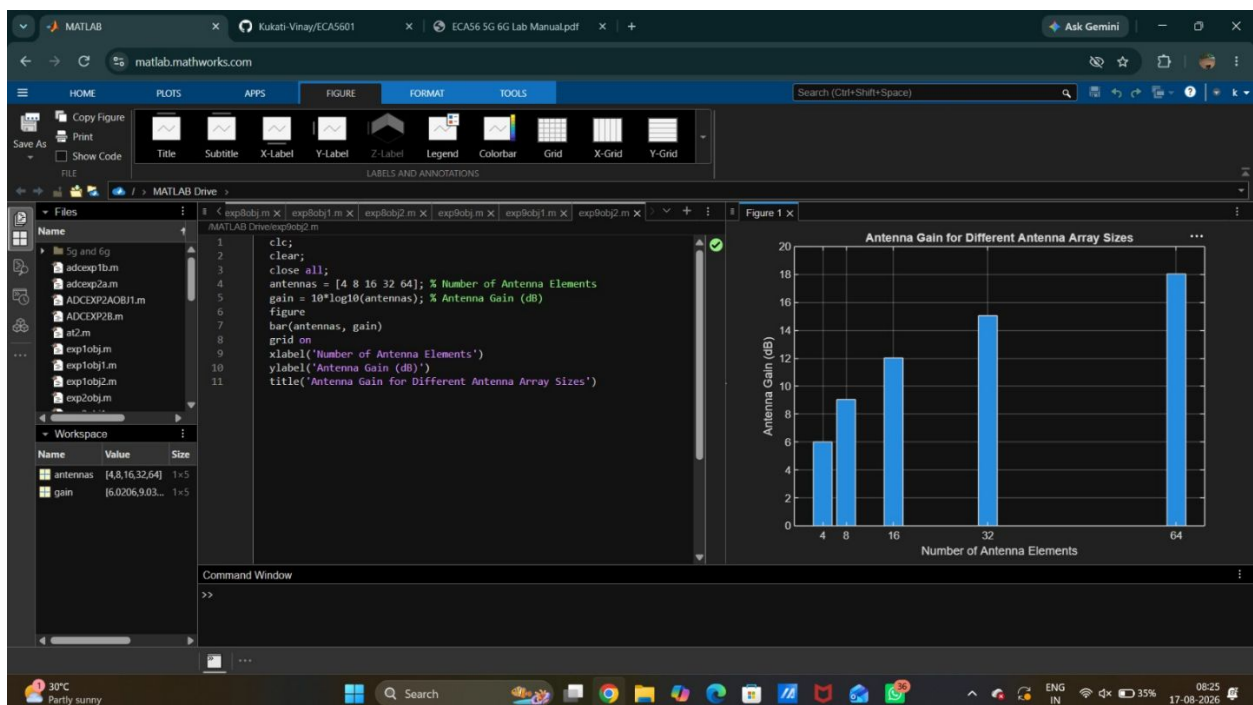
```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Antenna Gain (dB)')
```

```
title('Antenna Gain for Different Antenna Array Sizes')
```

output:



Result

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.